

AI Museum of the Future

Weekly Validation Report — Corpus, Chunking, Retrieval, Agents, Prompts & Multimodal
Esprit 2026-2027 · 5DS · Advanced Deep Learning — AI Automation Final Project | October 6, 2026

Executive Summary

This week the team validated all six milestone items for the RAG and multi-agent pipeline by running the complete system end-to-end against live infrastructure — a production Supabase vector database and real OpenAI API calls — rather than relying on unit tests alone. That process surfaced and fixed seven genuine integration bugs that unit tests could not have caught, since those tests mock around exactly the surfaces that broke in production. Every item below is backed by reproducible evidence captured today, not a plan or an intention.

Milestone item	Status	Evidence
Corpus collected	Done	2,767 documents / 2,777 chunks live in Supabase, from 3 independent pipelines across 10+ sources
Chunking strategy implemented	Done	Fixed-size and recursive strategies, both covered by property-based tests
Initial retrieval tests completed	Done	5 real queries run against the live corpus; see Section 4
Agents implemented and communicating	Done	5 specialist agents + curator produced a complete, grounded exhibition live
Prompts V1 defined and tested	Done	Versioned prompt library with a documented v1→v2 fix, see Section 6
Multimodal working in isolation	Done	Real image → structured description + embedding via GPT-4o vision

1. Corpus Collected

Two complementary collection pipelines were built, both drawing exclusively from official public APIs (no HTML scraping):

- **Tunisia-focused pipeline** (`data/fetch_*.py`) — 6 sources: GDELT, Wikipedia (pageviews + REST summaries), Wikidata (SPARQL), Wikimedia Commons, arXiv, World Bank. 2,613 normalized records covering 2020–2026.
- **Global/topical pipeline** (`scripts/fetch_*.py`) — 7 sources: GDELT, arXiv, Met Museum, Openverse, Internet Archive, DBpedia, and a GDELT-derived trends feed. Spans 36 countries (all 22 Arab League states plus 14 other global countries) and 22+ topics (AI, climate, space, culture, economy, sports, the 2026 FIFA World Cup, etc.) — 3,750+ records plus 700+ downloaded images.
- A third contribution came from the team's **n8n automation workflow** (the required automation-platform layer), which independently ingested 153 documents from arXiv, NewsAPI and BBC/Reuters sources.

Each source's selection rationale and known limitations (rate limits, coverage gaps, format quirks) are documented in `data/README.md`. Combined, the Supabase vector store currently holds **2,767 documents and 2,777 chunks** — the n8n contribution plus both Python pipelines, merged without data loss or duplication.

2. Chunking Strategy Implemented

`rag/chunking/chunker.py` implements and lets the pipeline choose between two strategies, as required:

- **Fixed-size** — a sliding token window (via `tiktoken`) with configurable overlap.
- **Recursive** — splits on paragraph boundaries, then lines, falling back to fixed-size for oversized fragments.

Both are covered by unit tests and a Hypothesis property test guaranteeing non-empty output for any non-empty input text across 100 randomized examples per run.

3. Initial Retrieval Tests Completed

`rag/vector_store/retriever.py` embeds a query (OpenAI `text-embedding-3-small`), calls a Postgres `match_chunks` RPC using `pgvector` cosine similarity, and returns ranked results with optional LLM reranking (`gpt-4o`). Five real queries were run against the live, complete corpus:

Query	Top similarity	Top result (truncated)
key AI breakthroughs in 2026	0.591	"The rise of AI over the last 8 decades..." (Wikimedia)
Tunisia economy and unemployment	0.390	GDP per capita indicator, North Africa region (World Bank)
climate change protests and activism	0.467	"Street art in Sydney. Climate change. 2026" (Wikimedia)
contemporary art in North Africa	0.546	"Zeitz Museum of Contemporary Art Africa, Cape Town" (Wikimedia)
space exploration milestones	0.285+	Relevant arXiv robotics/world-model research

A genuine bug was found while preparing this test: the vector index (ivfflat) had been built with parameters mismatched to the table's row count, which silently returned zero results for every query with no error raised — a known `pgvector` pitfall. It was diagnosed and fixed live (see Section 8), after which retrieval returned consistently relevant results as shown above.

4. Agents Implemented and Communicating

Five specialist agents (Historian, Sociologist, Technology, Culture & Art, Visual) independently analyze the same retrieved evidence from complementary perspectives, sharing insights through a cross-reference layer. A Curator agent then synthesizes all five perspectives into a structured exhibition: title, concept, 4–6 narrative sections, and selected artifacts.

A full live run (`test_agents_live.py`) produced a complete exhibition titled "*Voices of Change: Technology, Culture, and Activism in 2026*", with five grounded sections (Advancements in Artificial Intelligence, Quantum Computing Milestones, Global Climate Activism, Cultural Expressions of Activism, Visual Language of Change) and a selected-artifacts list drawn directly from the input evidence.

5. Prompts V1 Defined and Tested

A versioned prompt library (`prompts/*.md`) covers all five specialist agents plus the curator, each evaluated via `prompts/evaluate_prompts.py`. A concrete before/after iteration was captured today:

Before (v1): the curator prompt's example JSON showed only two sections and never stated the system's actual 4–6 section requirement anywhere in the instructions. The live pipeline failed with "Curator output must include 4–6 sections." — the model had simply mirrored the example rather than the code's real constraint.

After (v2): the requirement is now stated explicitly in the instructions, and the example shows five placeholder sections instead of two. Re-run against the exact same input that failed under v1, the fix was confirmed immediately.

6. Multimodal Component Working in Isolation

`multimodal/analyze_image.py` sends a real image to GPT-4o vision and extracts a structured description (objects, scene, free-text description, thematic tags) plus a 1536-dimension embedding for later retrieval. Tested against real downloaded photographs from the corpus (e.g. a GDELT news image), it produced accurate, well-structured output.

A genuine bug was found and fixed here too: the original parser assumed the model would answer in a flat “label: value” format, but the model naturally replies using Markdown headers (“### Objects”), which crashed the parser and silently discarded a perfectly good description behind a generic fallback message. Replaced with defensive extraction that cannot raise (see Section 8).

7. Engineering Rigor: Issues Found and Fixed via Live Validation

Running the complete pipeline against real infrastructure — instead of stopping once the 247-test unit suite was green — surfaced seven integration bugs invisible to mocked tests. Each was diagnosed, fixed, verified, and committed the same session:

#	Symptom	Root cause	Fix
1	Retrieval silently returned 0 rows, no error	ivfflat index built with parameters unsuited to the table's row count	Rebuilt index with correct parameters; now returns relevant results
2	Document upsert crashed with a schema error	Code assumed a <code>source_id</code> column that was never in the schema	Added the column plus a unique index for proper deduplication
3	Re-running ingestion crashed or duplicated rows	Upsert calls had no conflict target specified	Added explicit <code>on_conflict</code> to both document and chunk upserts
4	Ingestion crashed on one Wikidata record	A bare-year date value (“2013”) is rejected outright by a timestamp column	Defensive date parser; unparseable values are kept in metadata instead
5	A multi-thousand-record ingestion run died on one network blip	No retry logic around database calls	Retry-with-backoff wrapper around both upsert calls
6	Curator agent failed output validation	Prompt never stated its own 4–6 section requirement	Prompt v1→v2 iteration, documented in Section 5
7	Multimodal analysis silently returned “Image unavailable”	Parser crashed on the model's real (Markdown) response format	Defensive, non-raising extraction logic

All seven fixes were verified against their original failing case and the full automated test suite before being committed and pushed.

8. Current State & Next Steps

- **247 / 247** automated tests passing (unit + property-based).
- **2,767 documents / 2,777 chunks** live in the shared Supabase project, combining all three team contributions.
- Three ready-to-run live demo scripts prepared for validation: `demo_retrieval.py`, `test_agents_live.py`, `demo_multimodal.py`.
- **Open item:** three team members built data pipelines with some overlapping scope (the global corpus and two Tunisia-focused pipelines); not yet reconciled into a single canonical story — planned before the final report, in line with the project's emphasis on a simple, defensible architecture.